Compton scattering 2

From the previous section we have

$$p_1 = \underset{e^- \longrightarrow}{\begin{pmatrix} E \\ 0 \\ 0 \\ \omega \end{pmatrix}} \qquad p_2 = \underset{\longleftarrow \gamma}{\begin{pmatrix} \omega \\ 0 \\ 0 \\ -\omega \end{pmatrix}} \qquad p_3 = \underset{\nearrow\, e^-}{\begin{pmatrix} E \\ \omega \sin\theta \cos\phi \\ \omega \sin\theta \sin\phi \\ \omega \cos\theta \end{pmatrix}} \qquad p_4 = \underset{\gamma\, \swarrow}{\begin{pmatrix} \omega \\ -\omega \sin\theta \cos\phi \\ -\omega \sin\theta \sin\phi \\ -\omega \cos\theta \end{pmatrix}}$$

and

$$\langle |\mathcal{M}|^2 \rangle = \frac{e^4}{4} \left(\frac{f_{11}}{(s-m^2)^2} + \frac{2f_{12}}{(s-m^2)(u-m^2)} + \frac{f_{22}}{(u-m^2)^2} \right)$$

where

$$\begin{aligned} f_{11} &= -8su + 24m^2 s + 8m^2 u + 8m^4 \\ f_{12} &= 8m^2 s + 8m^2 u + 16m^4 \\ f_{22} &= -8su + 8m^2 s + 24m^2 u + 8m^4 \end{aligned}$$

Compton scattering experiments are typically done in the lab frame where the electron is at rest. Define boost Λ for transforming momentum vectors to the lab frame.

$$\Lambda = \begin{pmatrix} E/m & 0 & 0 & -\omega/m \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \omega/m & 0 & 0 & -E/m \end{pmatrix}$$

The electron is at rest in the lab frame.

$$\Lambda p_1 = \begin{pmatrix} m \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Mandelstam variables are invariant under a boost.

$$\begin{aligned} s &= (p_1 + p_2)^2 = (\Lambda p_1 + \Lambda p_2)^2 \\ t &= (p_1 - p_3)^2 = (\Lambda p_1 - \Lambda p_3)^2 \\ u &= (p_1 - p_4)^2 = (\Lambda p_1 - \Lambda p_4)^2 \end{aligned}$$

In the lab frame, let ω_L be the angular frequency of the incident photon and let ω'_L be the angular frequency of the scattered photon.

$$\omega_L = \Lambda p_2 \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \frac{\omega^2}{m} + \frac{\omega E}{m}$$

$$\omega'_L = \Lambda p_4 \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \frac{\omega^2 \cos\theta}{m} + \frac{\omega E}{m}$$

By computer algebra we have

$$\begin{aligned} s &= m^2 + 2m\omega_L \\ t &= 2m(\omega'_L - \omega_L) \\ u &= m^2 - 2m\omega'_L \end{aligned}$$

Hence

$$\langle|\mathcal{M}|^2\rangle = 2e^4 \left[\frac{\omega_L}{\omega'_L} + \frac{\omega'_L}{\omega_L} + \left(\frac{m}{\omega_L} - \frac{m}{\omega'_L} + 1\right)^2 - 1\right]$$

Lab scattering angle θ_L is given by the Compton equation

$$\cos\theta_L = \frac{m}{\omega_L} - \frac{m}{\omega'_L} + 1$$

Hence

$$\langle|\mathcal{M}|^2\rangle = 2e^4 \left(\frac{\omega_L}{\omega'_L} + \frac{\omega'_L}{\omega_L} + \cos^2\theta_L - 1\right)$$

Substitute $-\sin^2\theta_L$ for $\cos^2\theta_L - 1$.

$$\langle|\mathcal{M}|^2\rangle = 2e^4 \left(\frac{\omega_L}{\omega'_L} + \frac{\omega'_L}{\omega_L} - \sin^2\theta_L\right)$$

For simplicity let us drop the L subscript from lab variables. From now on the symbols ω, ω', and θ will be lab frame variables.

The differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{e^4}\left(\frac{\omega'}{\omega}\right)^2 \frac{\langle|\mathcal{M}|^2\rangle}{4s}$$

For the lab frame we have

$$\langle|\mathcal{M}|^2\rangle = 2e^4 \left(\frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2\theta\right)$$

Hence

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2s}\left(\frac{\omega'}{\omega}\right)^2 \left(\frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2\theta\right)$$

For $\omega \ll m$ we can use the approximation $s = m^2$ and obtain the Klein-Nishina formula

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2m^2}\left(\frac{\omega'}{\omega}\right)^2 \left(\frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2\theta\right)$$

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